

Two PSID Objects on the Down-Payment Constraint

Built May 26, 2026

This note documents two PSID empirical objects that together discipline the collateral constraint in the model and provide reduced-form evidence that the constraint matters for fertility. The model object is

$$b_t \geq (1 - \phi) p_i H_k,$$

with calibrated $\phi = 0.80$, so 20% of the price of an owned home in liquid wealth. Ownership in the model unlocks family-size housing services $\bar{h}(1) > \bar{h}(0)$. The model therefore predicts (i) actual first-time buyers cluster at or below the 20% threshold, and (ii) renters whose liquid wealth falls short of the implied down payment for a typical owned home are disproportionately childless. The two objects below test these two predictions.

1 Object A: Bunching at the Collateral Frontier

Construction

For each PSID head observed in 1984–2019, identify the first survey year in which the head is an owner (`HOMEOWN` = 1) and was previously observed as a renter (`HOMEOWN` = 2). Require `YEARMOVED_` $\in [t - 2, t]$ so reported value and mortgage are at or near origination. The down-payment share at first purchase is

$$\frac{DP}{P} = 1 - \frac{HOMEMORTOTR_t}{HOMEVALUER_t}.$$

Cash buyers (`HOMEMORTOTR` = 0) are 17.9% of the sample and are shown separately. The headline density is plotted conditional on having a mortgage so the cash spike at $DP/P = 1$ does not visually distract from the bunching story.

Result

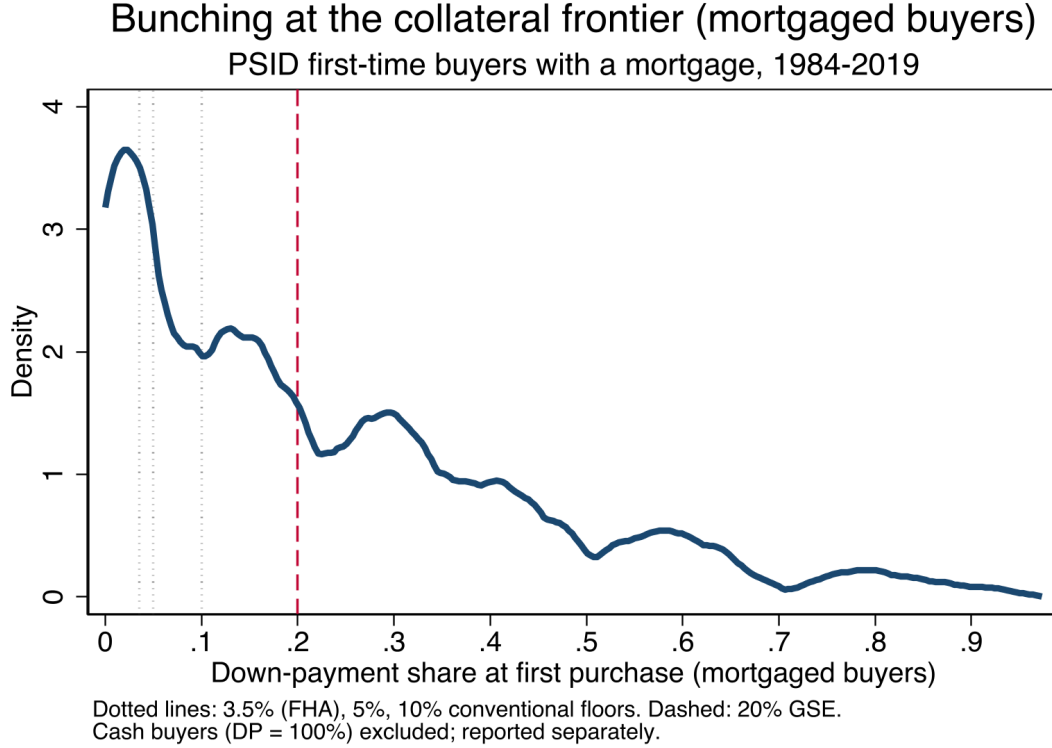


Figure 1: Kernel density of the down-payment share at first home purchase, PSID first-time buyers with a mortgage, 1984–2019. Dotted lines mark 3.5% (FHA), 5%, and 10% conventional floors; the dashed red line is the 20% GSE conforming threshold (the calibrated $1 - \phi$).

Table 1: Down-payment share at first purchase, mortgaged buyers.

Statistic	Value
First-time mortgaged buyers (N)	252
Median DP/P	0.16
Share with $DP/P \leq 0.035$ (FHA)	0.241
Share with $DP/P \leq 0.05$	0.282
Share with $DP/P \leq 0.10$	0.378
Share with $DP/P \leq 0.20$ (GSE)	0.566
Cash buyers (separate)	0.179

The density is monotone decreasing from 0 to 0.20 with a visible step right at the 20% GSE line and a thin right tail. The bunching is not a 2008–2010 crisis artifact: the share at or below 20% is 47% in 1995–2004, 43% in 2005–2010, and 50% in 2011–2019 (raw shares including cash buyers; the mortgaged-only versions are higher).

What Object A buys

Direct, model-free evidence that real first-time buyers operate at the collateral frontier the model writes down. The DP constraint is where the modal mortgaged first-time buyer lands.

2 Object B: DP Shortfall Predicts Childlessness

Construction

Build a per-year target home price as the median value among observed first-time PSID buyers in the same year-bin (Table 2). For each PSID head observed as a renter at ages 25–30, define

$$\text{shortfall}_{it} = \mathbf{1}\left\{b_{it} < 0.20 \cdot P_t^{\text{target}}\right\},$$

where $b_{it} = \text{NETWORTH2R}$ (PSID net worth excluding home equity). Collapse to one observation per ID (mean over the 25–30 window), keep observations followed to age 45, and compare childlessness by 45 across shortfall status within income tercile.

Table 2: Target home price P_t^{target} by year-bin (PSID first-time buyer median home value, real).

Year bin	P^{target} (real \$)	First-time buyers
1995–2004	179,300	95
2005–2010	188,844	128
2011–2019	218,775	169

Result

Renters who lack the 20% down payment remain childless more
PSID renters at 25-30, pre-first-birth, by DP shortfall and income tercile

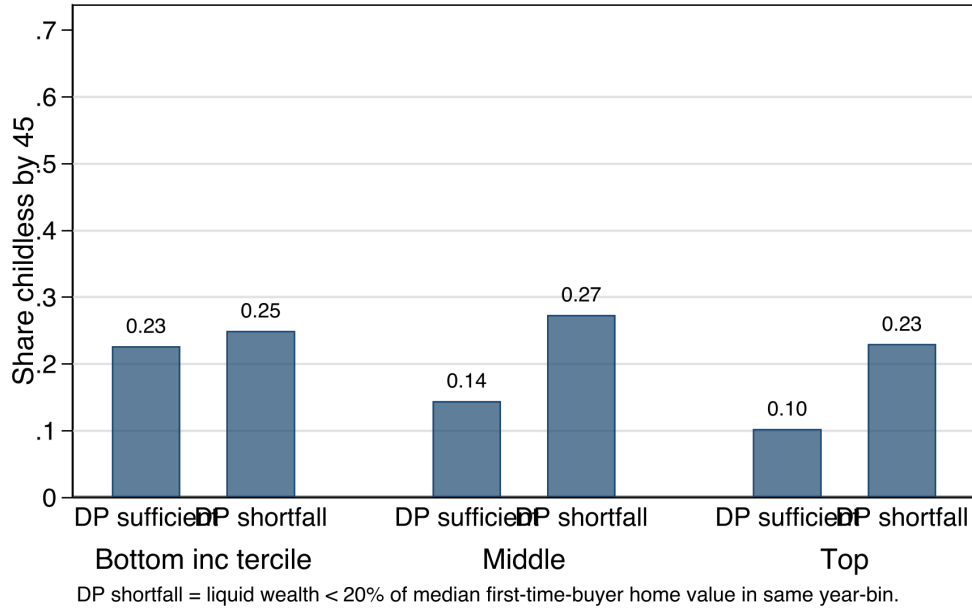


Figure 2: Share childless by 45 among PSID renters at ages 25–30, split by DP-shortfall status within income tercile.

Table 3: Childlessness by 45 by shortfall status and income tercile (PSID renters at 25–30).

Income tercile	DP sufficient	<i>N</i>	DP shortfall	<i>N</i>	Gap
Bottom	0.227	19	0.250	1,237	+2.3 pp
Middle	0.145	37	0.274	864	+ 12.9 pp
Top	0.103	59	0.230	758	+ 12.7 pp

Table 4: OLS of childless-by-45 on DP shortfall and controls. PSID renters at 25–30, pre-first-birth, $N = 2,961$. PSID weights, robust SEs.

Specification	$\hat{\beta}_{\text{shortfall}}$	SE	<i>p</i>
Shortfall (20%)	0.115	0.039	0.003
Shortfall (20%) + log income	0.116	0.040	0.003
Shortfall (20%) + log income + log NW	0.122	0.041	0.003
Strict shortfall (10%) + log income	0.068	0.036	0.063

DP-shortfall renters are about 12 pp more childless by 45, controlling for log income and log net worth. The 10% strict threshold is weaker, consistent with 20% being the binding margin in both this object and the calibrated ϕ .

What Object B buys

Direct reduced-form evidence that the constraint matters for fertility, not just for tenure. The threshold is sharper than raw wealth because it varies with both wealth and prices (the threshold $0.20 \cdot P_t^{\text{target}}$ moves over time), and the within-income split nets out the income channel.

3 Why Both Together Make the Case

Object A alone says the constraint binds for buyers; a critic can respond that non-buyers stay renters for reasons unrelated to fertility. Object B alone says renters with little wealth have fewer kids; a critic can respond that the result is just income or wealth. Together, the *same* 20% threshold that buyers bunch at also separates childless from parent renters within income tercile. Two complementary moments of the same distribution, both consistent with the calibrated ϕ .

4 Draft Pitch Frames

These are drafts to paste into `latex/may_29_project_presentation.tex`. Style matches the existing PSID frames around lines 1188 and 1235. Adjust hyperref targets if needed.

Frame 1: Bunching at the collateral frontier

```
\begin{frame}{Buyers Bunch at the 20\% Down-Payment Constraint}
\begin{columns}[T]
\begin{column}{0.58\textwidth}
\centering
\includegraphics[width=\linewidth]{../code/data/psid_followup_mar2026/output/
    dp_at_first_purchase_v1/dp_share_density_nocash_v1.png}
\end{column}
\begin{column}{0.40\textwidth}
\small
PSID first-time buyers with a mortgage, $1984$--$2019$ ($N=252$).
\medskip
\begin{itemize}
\item  $\text{DP}/P = 1 - \text{mortgage}/\text{value}$  at first owner-year.
\item Density is monotone from $0$ to $0.20$ with a visible step at the GSE 20\% line.
\item  $\mathbf{57\%}$  of mortgaged buyers put down  $\leq 20\%$ ; 38\% put down  $\leq 10\%$ .
\item Stable across $1995$--$2004$, $2005$--$2010$, $2011$--$2019$: not a crisis artifact.
\end{itemize}
\end{column}
\end{columns}

\medskip
\footnotesize The calibrated  $\phi = 0.80$  matches the threshold the modal mortgaged first-time buyer hits in the data.
\end{frame}
```

Frame 2: Down-payment shortfall predicts childlessness

```
\begin{frame}{Renters Below the Down-Payment Frontier Stay Childless}
\begin{columns}[T]
\begin{column}{0.58\textwidth}
```

```

\centering
\includegraphics[width=\linewidth]{../code/data/psid_followup_mar2026/output/
    dp_shortfall_v1/childless_by_shortfall_inc_tercile_v1.png}
\end{column}
\begin{column}{0.40\textwidth}
\small
PSID renters at $25$--$30$, pre-first-birth, $N=2\{,\}961$.
\medskip
\begin{itemize}
\item Shortfall  $\$ = \$$  liquid wealth  $\$ < 0.20 \times \$$  median first-time-buyer home
    value in that year-bin.
\item Within \emph{every} income tercile, shortfall renters are more childless by
    $45$.
\item Mid- and top-income terciles: gap is  $\$ \mathbf{\approx 13} \$$  pp.
\item Regression coefficient  $\$0.122\$$  (SE  $\$0.041\$$ ,  $\$p=0.003\$$ ), controlling for  $\$ \log \$$ 
    income and  $\$ \log \$$  net worth.
\end{itemize}
\end{column}
\end{columns}

\medskip
\footnotesize The 20\% threshold is the same number that buyers bunch at and the same
    number the model calibrates as  $\$1 - \phi\$$ .
\end{frame}

```

5 Caveats

- **Sample sizes:** PSID is small. Object A has $N = 252$ mortgaged first-time buyers, Object B has $N = 2,961$ pre-birth renters followed to 45. Within bottom income tercile, the DP-sufficient comparison group in B has only $N = 19$, so that row should be read as descriptive only.
- **Origination proxy:** DP/P is computed from current HOMEVALUER and current HOMEMORTOTR at the first surveyed owner-year, with $YEARMOVED_ \in [t - 2, t]$. Some drift from true origination is unavoidable. House-price appreciation post-purchase biases reported DP/P *up*, so the bunching headline is conservative.
- **Gifts and transfers:** A buyer financed by a family gift looks like a high-DP buyer; this also biases the bunching estimate down. The true share at the collateral floor is plausibly higher than reported.
- **Target price in B:** uses national year-bin median first-time-buyer value. A state- or metro-level analog would tighten the threshold; the PSID shelf has STATEFIPS_ but not MSA. State-level version is straightforward as a robustness pass.
- **Outright cash buyers** are 17.9% of A's sample. They are excluded from the headline density and reported separately; the all-buyers density and summary are still saved as `dp_share_density_v1.png` and `dp_share_bunching_summary_v1.csv`.

6 Code and Artifacts

- Do-files: `code/data/psid_followup_mar2026/dp_at_first_purchase_v1.do` (Object A), `code/data/psid_followup_mar2026/dp_at_first_purchase_v1.do` (Object B).
- Outputs: `code/data/psid_followup_mar2026/output/dp_at_first_purchase_v1/` and `code/data/psid_followup_mar2026/output/dp_at_first_purchase_v1/`